KL Divergence

(Kullback-Leibler)

KL divergence

- Measure of "distance" between two distributions P(i) and Q(i)

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$: discrete pdf

$$D_{KL}(P||Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx$$: continuous pdf

- we want to quantify how similar two probability distributions.

(Fair)

Coin 1

$$\begin{cases} 0.5 & heads \\ 0.5 & tails \end{cases}$$

(Biased)

Coin 2

$$\begin{cases} p & heads \\ q & tails \end{cases}$$

We want to measure how similar these two distributions are. (e.g. we want to understand whether coin 2 is biased or not)

quite similar distributions

Coin 1

$$\begin{cases} 0.5 & heads \\ 0.5 & tails \end{cases}$$

Coin 2

$$\begin{cases} 0.55 & heads \\ 0.45 & tails \end{cases}$$

H H T H H T T H T H T H

H H T H H T T H H H T H

- If you flip both coins many times distribution may be confused.

Coin 1

$$\begin{cases} 0.5 & heads \\ 0.5 & tails \end{cases}$$

Coin 2

$$\begin{cases} 0.95 & heads \\ 0.05 & tails \end{cases}$$

H H T H H T T H T H T H

H H H H H H T H H H H H

- will be easier to identify that the two distributions are different by observation.

Coin 1

$$\begin{cases} 0.5 & heads \\ 0.5 & tails \end{cases}$$

Coin 2

$$\begin{cases} 0.95 & heads \\ 0.05 & tails \end{cases}$$

A quantitative measure of similarity:

$$\frac{P\,(\text{Observations} \mid \text{real coin})}{P\,(\text{Observations} \mid \text{coin 2})}$$

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Flip the coin N times and record the obsrvations.

H H T H H T H H H T H T

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1$$

(Calculate the probability of the observed sequence.)

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1$$

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1 \cdot p_1 \cdot p_2 \cdot p_1 \cdot p_1$$

$$q_1 \cdot q_1 \cdot q_2 \cdot q_1 \cdot q_1 \cdot q_2 \cdot q_1 \cdot q_1 \cdot q_1 \cdot q_2 \cdot q_1 \cdot q_2$$

write the probability of the same sequence of obsevations being generated by the second coin

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

P (Observations | real coin) $= p_1^{N_H} p_2^{N_T}$

P (Observations | coin 2) $q_1^{N_H} q_2^{N_T}$

N_H : # of Heads

N_T : # of tails

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T : sequence

$$\frac{P \text{ (Observations | real coin)}}{P \text{ (Observations | coin 2)}} = \frac{p_1^{N_H} p_2^{N_T}}{q_1^{N_H} q_2^{N_T}}$$

(Find the ratio of probabilities)

- The likelihood of the sequence having been generated by the real coin.
- The KL divergence measures the exact same thing

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$\left(\frac{p_1^{N_H} \, p_2^{N_T}}{q_1^{N_H} \, q_2^{N_T}} \right)^{\frac{1}{N}}$$

Raise the ratio to the power of $\frac{1}{N}$

N : # samples

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$\log \left(\frac{p_1^{N_H} \, p_2^{N_T}}{q_1^{N_H} \, q_2^{N_T}} \right)^{\frac{1}{N}}$$

take the logarithm

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$\frac{1}{N} \log \left(\frac{p_1^{N_H} \, p_2^{N_T}}{q_1^{N_H} \, q_2^{N_T}} \right)$$

The exponent of $\frac{1}{N}$ can be written as a multiplication.

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$\frac{1}{N}\log p_1^{N_H} + \frac{1}{N}\log p_2^{N_T} \; - \; \frac{1}{N}\log q_1^{N_H} - \frac{1}{N}\log q_2^{N_T}$$

- write multiplications and divisions as additions and subtractions.

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$\frac{N_H}{N}\log p_1 + \frac{N_T}{N}\log p_2 - \frac{N_H}{N}\log q_1 - \frac{N_T}{N}\log q_2$$

(Drop down the powers)

- The sequence was generated by the True coin.

Hence as $N \to \infty$, $\frac{N_H}{N} \to p_1$, $\frac{N_T}{N} \to p_2$ in the limit.

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$p_1 \log p_1 + p_2 \log p_2 - p_1 \log q_1 - p_2 \log q_2$$

Logarithmic manipulations:

$$p_1 \log p_1 - p_1 \log q_1 + p_2 \log p_2 - p_2 \log q_2$$

$$p_1 \log \frac{p_1}{q_1} + p_2 \log \frac{p_2}{q_2}$$

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$p_1 \log \frac{p_1}{q_1} + p_2 \log \frac{p_2}{q_2}$$

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$p_1 \log \frac{p_1}{q_1} + p_2 \log \frac{p_2}{q_2}$$

$$\text{Log} \left(\frac{P(\text{Observations of d1} \mid \text{distribution 1})}{P(\text{Observations of d1} \mid \text{distribution 2})} \right)$$

*d1 = distribution 1

True Coin

$$\begin{cases} p_1 & heads \\ p_2 & tails \end{cases}$$

Coin 2

$$\begin{cases} q_1 & heads \\ q_2 & tails \end{cases}$$

H H T H H T H H H T H T

$$D_{KL}(P||Q) = \sum_i P(i)\log\frac{P(i)}{Q(i)}$$

$$p_1\log\frac{p_1}{q_1} + p_2\log\frac{p_2}{q_2}$$

$$\text{Log}\left(\frac{P(\text{Observations of d1} \mid \text{distribution 1})}{P(\text{Observations of d1} \mid \text{distribution 2})}\right)$$

*d1 = distribution 1

equivalent

$D_{KL}(P||Q)$ looks at how likely the second distribution will be able to generate the samples from the first distribution

- In machine learning KL divergence is used in many places.
  - e.g. The cross-entropy loss is equivalent to the KL loss. By minimizing cross-entropy we're minimizing the distance between distributions.

KL divergence

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

$$D_{KL}(P||Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx$$

Reference

- Adian Liusie
- https://www.youtube.com/watch?v=SxGYPqCgJWM